AI-Based Student Sports Talent & Fitness Guidance Portal

Final Project Report for UCS503P
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1 Higher-order goal

. . . secondary framing

This project aims to help schools move from ad-hoc, paper-based fitness records toward evidence-informed sports guidance, so that students with genuine athletic potential are less likely to be overlooked and students who are interested in sport receive individualised, data-backed direction rather than generic advice. The broader framing – more capable, confident young athletes – is treated as a secondary, motivating lens; the primary engineering commitment remains a working, coach-supervised assessment-to-guidance pipeline.

2 Time-to-value

. . . primary heuristic

- We delivered the core assessment-to-prediction workflow first, before any secondary feature (reporting, dashboards), so that the riskiest technical question – can a fitness-level model be trained honestly on available data – was answered early.
- We prioritised a working, if narrow, vertical slice (assessment entry → ML prediction → coach approval → student visibility) over a wide but shallow feature set.
- Success for this stage is defined by what is demonstrably working end-to-end today, documented honestly in Section 9, rather than by what was originally planned in the proposal.

3 Problem Statement

Schools conduct sports activities regularly, but student fitness and performance data is rarely recorded in a structured, comparable way. This creates three concrete problems, unchanged since the proposal stage and confirmed during development:

- **Missed potential:** students with genuine athletic ability can go unnoticed without systematic, school-wide comparison of fitness data.
- **Generic guidance:** interested students rarely receive individualised direction on which sport suits their physical profile or what to improve.
- **No progress visibility:** without repeated, comparable assessments, neither the student nor the coach can tell whether training is working.

Why this matters now: the Khelo India fitness-test framework already gives schools a standard, low-cost battery of tests; the missing piece is a system that turns repeated test results into structured, coach-reviewed guidance rather than a one-time paper record.

4 Proposed Solution *. . . Software Proposition*

4.1 Overview

The delivered system is a **web-based** sports talent and fitness guidance portal with the following components:

- **Coach assessment entry:** a coach records a student's general fitness test results (strength, endurance, flexibility, speed-agility, BMI, attendance, motivation).
- **ML-assisted fitness classification:** a trained classifier predicts a fitness category and a confidence score from the recorded assessment.
- **Coach approval workflow:** every prediction is held back from students and parents until a coach explicitly approves or overrides it.
- **Student/parent guidance view:** approved results, together with assessment history, are shown to the student and, at summary level, to the parent.
- **PDF report generation:** a downloadable summary report per student.

4.2 Core workflow (as implemented)

1. Coach submits a fitness assessment for a student through the web frontend.
2. Backend validates the submission and stores it.
3. Backend invokes the Python ML service, which returns a predicted fitness level and confidence score.
4. Coach reviews the prediction on the coach dashboard and marks it *approved* or *override*.
5. Only approved results become visible in the student and parent views, together with historical assessments for that student.

4.3 Deliberate scope narrowing: two-tier design

Early in development we identified a concrete risk: collecting every conceivable metric for every possible sport, for every student, from day one, would make the system too complex for a coach to realistically operate. We therefore split the data model into two tiers: **Tier 1 – General Athleticism**, a single universal fitness battery applied to every student and used to recommend suitable sports, which is the complete scope of the current semester; and **Tier 2 – Sport-Specific KPIs**, a small, per-sport indicator set applied only to students who are funnelled into a specific sport squad, explicitly deferred to future work (Section 9).

5 Solution Approach *. . . Engineering focus*

As an engineering course project, the emphasis throughout was on correctness, honest evaluation, and workflow design, rather than on novel algorithms.

5.1 Fitness classification *. . . non-research*

Fitness-level prediction uses standard, explainable models rather than research-grade architectures:

- A Random Forest classifier, trained with SMOTE oversampling to address class imbalance, chosen specifically because its decisions can be inspected and explained to a non-technical coach.
- A leakage-control step was applied before any accuracy figure was trusted: an initial model that included a derived "overall performance" column reached a suspicious 98% accuracy by relying on that column instead of genuine fitness signals; removing such columns produced a more credible, believable model.
- The model is shown to the user only as decision support: a coach must approve or override every prediction before it is acted on.

5.2 Web architecture *... web-based solution*

A 3-tier web architecture, shown in Figure 1:

- **Frontend:** React/Vite interface for coach, student, parent, and admin views.
- **Backend API:** Node.js/Express, handling validation, persistence, and orchestration of the ML service.
- **ML service:** a Python/scikit-learn prediction script invoked by the backend per assessment.
- **Data store:** designed around a normalised PostgreSQL schema (Section 7); the current implementation persists assessments in a flat structure as an interim step.

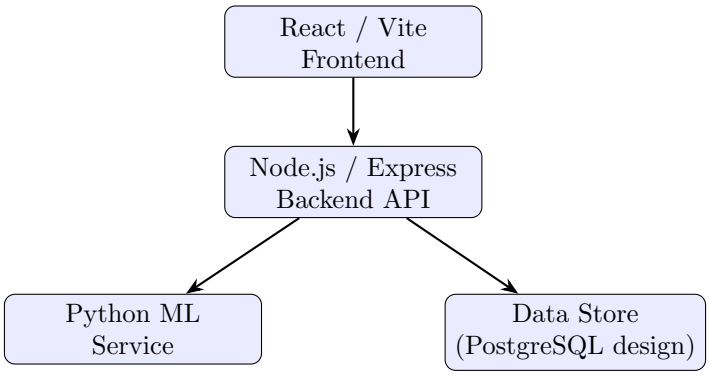


Figure 1: Implemented system architecture

5.3 Iteration and delivery

Development proceeded in short, working increments: the assessment-entry and BMI-calculation module was completed and tested before the ML service was integrated; the ML service was validated in isolation (including the leakage-detection exercise above) before being wired into the backend; and the coach-approval gate was added and tested before student/parent views were connected to it. This ordering meant that at every stage there was a working, demonstrable slice of the system, rather than several half-finished components.

6 Evaluation Criterion *... measurable and attributable*

As proposed, we define success using measurable, attributable metrics tied to the core workflow, and we report the actual measured values rather than targets alone.

6.1 Primary evaluation metric

Classification quality on a held-out test set.

- **Result:** 83% overall accuracy, macro-averaged F1-score of 0.56, on a 100-record held-out test split (80/20 split of 500 records).
- **Attribution:** measured directly via `scikit-learn`'s classification report and confusion matrix on the trained pipeline; figures were independently re-run and confirmed rather than taken on trust from an earlier draft.
- **Honest limitation:** the High Performer class (12 of 500 records) was correctly identified 0 times in the test set, even with SMOTE and balanced class weights – a genuine data-availability limitation, reported here rather than masked by the relatively high overall accuracy.

6.2 Secondary evaluation metrics

- **Workflow integrity:** a prediction is verified to be inaccessible to the student/parent view until a coach approval action is recorded, tested via an end-to-end integration test of Assessment → ML → Approval → Visibility.
- **Role isolation:** each of the four roles (Student, Coach, Admin, Parent) was verified to access only its intended views.
- **Explainability:** every stored prediction carries its confidence score, so no recommendation is presented as an unexplained black-box output.

6.3 Pilot validation plan

- A small-scale internal test used synthetic and openly available student fitness records rather than real school data, consistent with the project's data-privacy constraints for a student prototype.
- Coach-facing usability (can a coach complete an assessment and review a recommendation without guidance) was checked informally during development and is proposed as the basis for a future formal pilot once real coach users are available.

7 Scalability *... with foresight*

1. **Scalable (theoretically):** the system is designed around a normalised relational schema (`Students`, `Assessments`, `Coaches`, `Recommendations`, `Admins`) so that a school-wide cohort, assessed repeatedly over time, can be queried efficiently rather than requiring a redesign as data volume grows. The schema is deliberately additive: a future `Sport_Specific_Metrics` table for Tier 2 can be introduced without modifying the Tier 1 tables.
2. **Operationally deployable (practically):** the frontend, backend, and ML service are decoupled and independently deployable (e.g. Vercel for the frontend, a Node-compatible host for the backend), and the ML service is invoked as a stateless per-request call, avoiding a persistent, hard-to-scale ML process.

8 Engine availability heuristic

This project deliberately relies on mature, off-the-shelf engines rather than custom infrastructure, consistent with the course's non-research framing:

- **Web framework:** Express (REST API) and React (frontend), both mature, widely documented.
- **ML library:** scikit-learn and imbalanced-learn, rather than a custom-built model or training loop.
- **Database engine:** PostgreSQL, a mature relational engine, chosen specifically because the data is inherently relational (repeated assessments per student).
- **Report generation:** standard client-side PDF generation rather than a custom document-rendering pipeline.

Using these mature components let the team focus on workflow correctness, leakage-aware evaluation, and the coach-approval design, rather than on infrastructure that already exists in well-tested form elsewhere.

9 Project scope and deliverables *... incremental, results over intentions*

9.1 Delivered in this stage

- Coach assessment entry with automatic BMI calculation.
- Trained and evaluated fitness-level classifier (Random Forest with SMOTE), including a documented leakage-detection and correction process.
- Coach approval/override workflow, enforced before student or parent visibility.
- Student dashboard showing approved recommendations and assessment history.
- Parent dashboard showing a summary-level, coach-approved view.
- Downloadable PDF fitness report.

9.2 Not yet delivered, and why

In the spirit of reporting results rather than intentions, the following gaps are stated plainly rather than implied as complete:

- **Sport-suitability recommendation logic** (mapping a fitness profile to specific suggested sports) is specified but not yet implemented; the current system predicts a fitness category only.
- **Relational database migration:** assessments are currently persisted in a flat structure rather than the normalised PostgreSQL schema described in Section 7; this is the next planned increment.

9.3 Subsequent deliverables (Tier 2 and beyond)

- Sport-specific KPI tracking for squad-selected students, added one sport at a time.
- A progression/forecasting model comparing a student's improvement trajectory against age-group benchmarks.
- A larger, more representative dataset, particularly to make the High Performer category reliably predictable.

10 Risks and mitigations

- **Overtrusting model accuracy:** mitigated by the explicit leakage-detection exercise (Section 5.1) and by reporting per-class, not just overall, metrics.
- **Data privacy / consent:** mitigated by using only synthetic or openly available fitness data during development; no real student-identifying data was collected.
- **Coach adoption:** mitigated by keeping assessment entry to a single form with automatic BMI calculation, and by never surfacing an unreviewed prediction to a student or parent.
- **Scope creep from sport-specific tracking:** mitigated directly by the two-tier design (Section 4.3), which confines the current semester to a single, universal fitness battery.

11 Summary

This report documents the transition of the AI-Based Student Sports Talent & Fitness Guidance Portal from a proposal to a working system. The core assessment-to- guidance workflow – coach entry, ML-assisted fitness classification, mandatory coach approval, and student/parent visibility – is implemented and tested end to end, and the underlying classifier was evaluated honestly, including a documented leakage-detection process and an explicit acknowledgement of its class-imbalance limitation. Sport-specific recommendation logic and full relational persistence remain the immediate next steps, and are reported here as open work rather than implied as complete. The project continues to meet the course's selection criteria by favouring working increments over a wide but shallow feature set, using mature, off-the-shelf engines rather than research-grade components, and defining success through measurable, attributable evaluation criteria rather than subjective impressions of progress.